```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Neuro-Quantum Brain | Trading Console</title>
  <!-- Tailwind CSS -->
  <script src="https://cdn.tailwindcss.com"></script>
  <!-- Chart.js -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <!-- Phosphor Icons -->
  <script src="https://unpkg.com/@phosphor-icons/web"></script>

  <!-- Chosen Palette: Modern Fintech Light Theme
     Background: Slate-50/White
     Accents: Emerald-600 (Buy), Rose-600 (Sell), Indigo-600 (Tech/AI)
     Text: Slate-800
  -->

  <!-- Application Structure Plan:
     1. Header: System Status & Global Metrics.
     2. Main Grid:
       - Left Col: "The Neural Core" (Logic Interactive). Users manually adjust Hurst/Entropy inputs to test the PDF's 'Secret Formula'.
       - Center Col: "Main Battlefield" (Dynamic Charts). Tabs to switch between Kalman, Fisher, and Smart Indicators.
       - Right Col: "War Room Log". A scrolling terminal showing system messages in Tamil/English.
     3. Footer: System Integrity check.
```

Rational: This structure separates the "Math/Logic" (Left) from the "Visual Result" (Center), allowing users to understand the 'Why' before seeing the 'What'.

-->

<!-- Visualization & Content Choices:

- Neural Logic: HTML Range Sliders + JS Logic. Goal: Teach the 'Secret Formula' interaction.

- Trend/Kalman: Line Chart (Chart.js). Goal: Show smooth trend tracking vs noisy price.

- Fisher: Bar/Line Combo (Chart.js). Goal: Visualize reversal points (-2 to +2).

- Terminology: Strictly using the mixed Tamil/English phrasing from source PDFs (e.g., "மரண மாஸ் என்ட்ரி").

-->

<!-- CONFIRMATION: NO SVG graphics used. NO Mermaid JS used. -->

<style>

@import
```
url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=Noto+Sans+Tamil:wght@400;500;700&display=swap');
```

```
body {
  font-family: 'Inter', 'Noto Sans Tamil', sans-serif;
  background-color: #f8fafc; /* Slate 50 */
}

.chart-container {
  position: relative;
  width: 100%;
```

```
        height: 350px;
        max-height: 400px;
      }

      .toggle-checkbox:checked {
        right: 0;
        border-color: #4f46e5;
      }
      .toggle-checkbox:checked + .toggle-label {
        background-color: #4f46e5;
      }

      /* Custom Scrollbar for Log */
      .console-log::-webkit-scrollbar {
        width: 6px;
      }
      .console-log::-webkit-scrollbar-track {
        background: #f1f5f9;
      }
      .console-log::-webkit-scrollbar-thumb {
        background: #cbd5e1;
        border-radius: 3px;
      }
    </style>
  </head>
  <body class="text-slate-800">

    <!-- Navbar -->
```

```
<nav class="bg-white border-b border-slate-200 shadow-sm sticky top-0 z-50">
  <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
    <div class="flex justify-between h-16">
      <div class="flex items-center gap-3">
        <div class="bg-indigo-600 text-white p-2 rounded-lg">
          <i class="ph ph-brain text-2xl"></i>
        </div>
        <div>
          <h1 class="font-bold text-xl tracking-tight text-slate-900">Neuro-Quantum Brain</h1>
          <p class="text-xs text-slate-500 font-medium">AI DRIVEN TRADING SYSTEM (தமிழ் பதிப்பு)</p>
        </div>
      </div>
      <div class="flex items-center gap-4">
        <div class="hidden md:flex flex-col items-end mr-4">
          <span class="text-xs font-bold text-emerald-600 flex items-center gap-1">
            <span class="w-2 h-2 rounded-full bg-emerald-500 animate-pulse"></span> SYSTEM ONLINE
          </span>
          <span class="text-xs text-slate-400">v4.2.0 (Stable)</span>
        </div>
      </div>
    </div>
  </div>
</nav>

<!-- Main Dashboard Container -->
<div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">
```

```
<!-- Intro Section -->
<div class="mb-8">
  <h2 class="text-2xl font-bold text-slate-900 mb-2">War Room Console (கட்டளை மையம்)</h2>
  <p class="text-slate-600 max-w-3xl">
    இந்த டேஷ்போர்டு உங்களின் அனைத்து AI மாடல்களையும் ஒருங்கிணைக்கிறது.
    Kalman Trend, Smart Indicators, மற்றும் Fisher Transform அல்காரிதம்களின் நேரடி செயல்பாட்டை இங்கே கண்காணிக்கலாம்.
    (This dashboard integrates all your AI models to monitor live performance.)
  </p>
</div>

<!-- Top Metrics Row -->
<div class="grid grid-cols-1 md:grid-cols-4 gap-4 mb-8">
  <!-- Metric 1 -->
  <div class="bg-white p-4 rounded-xl border border-slate-200 shadow-sm flex items-center justify-between">
    <div>
      <p class="text-xs text-slate-500 uppercase font-semibold">Nifty 50 Trend</p>
      <h3 class="text-xl font-bold text-emerald-600 mt-1">BULLISH</h3>
      <p class="text-xs text-slate-400 mt-1">Confidence: 85%</p>
    </div>
    <div class="bg-emerald-50 p-2 rounded-full text-emerald-600">
      <i class="ph ph-trend-up text-xl"></i>
    </div>
  </div>
  <!-- Metric 2 -->
```

```
<div class="bg-white p-4 rounded-xl border border-slate-200 shadow-sm flex items-center justify-between">
  <div>
    <p class="text-xs text-slate-500 uppercase font-semibold">Bank Nifty Volatility</p>
    <h3 class="text-xl font-bold text-amber-600 mt-1">MODERATE</h3>
    <p class="text-xs text-slate-400 mt-1">VIX: 14.2</p>
  </div>
  <div class="bg-amber-50 p-2 rounded-full text-amber-600">
    <i class="ph ph-activity text-xl"></i>
  </div>
</div>
<!-- Metric 3 -->
<div class="bg-white p-4 rounded-xl border border-slate-200 shadow-sm flex items-center justify-between">
  <div>
    <p class="text-xs text-slate-500 uppercase font-semibold">AI Decision (Current)</p>
    <h3 class="text-xl font-bold text-indigo-600 mt-1">WAIT</h3>
    <p class="text-xs text-slate-400 mt-1">Scanning for Setup...</p>
  </div>
  <div class="bg-indigo-50 p-2 rounded-full text-indigo-600">
    <i class="ph ph-robot text-xl"></i>
  </div>
</div>
<!-- Metric 4 -->
<div class="bg-white p-4 rounded-xl border border-slate-200 shadow-sm flex items-center justify-between">
  <div>
```

```
            <p class="text-xs text-slate-500 uppercase font-semibold">Trap
Detection</p>
            <h3 class="text-xl font-bold text-slate-700 mt-1">SAFE ZONE</h3>
            <p class="text-xs text-slate-400 mt-1">Entropy < 0.5</p>
          </div>
          <div class="bg-slate-100 p-2 rounded-full text-slate-600">
            <i class="ph ph-shield-check text-xl"></i>
          </div>
        </div>
      </div>

      <!-- Main Content Grid -->
      <div class="grid grid-cols-1 lg:grid-cols-12 gap-8">

        <!-- LEFT COLUMN: The Brain Logic (Interactive) -->
        <div class="lg:col-span-4 space-y-6">

          <!-- Logic Controller Card -->
          <div class="bg-white rounded-xl border border-slate-200 shadow-sm overflow-
hidden">
            <div class="bg-slate-50 px-6 py-4 border-b border-slate-200">
              <h3 class="font-bold text-slate-800 flex items-center gap-2">
                <i class="ph ph-cpu text-indigo-600"></i> The Secret Formula (Logic Core)
              </h3>
            </div>
            <div class="p-6">
              <p class="text-sm text-slate-600 mb-6 leading-relaxed">
                "இரண்டு மூளையும் பேசி முடிவு எடுக்கும் இடம்." <br>
```

AI Buy சொன்னாலும், Hurst (Trend) மற்றும் Entropy (Chaos) சரியாக இருந்தால் மட்டுமே அது "Killer Trade".

</p>

<!-- Interactive Sliders -->

<!-- Neural Signal Input -->

<label class="block text-xs font-bold text-slate-500 uppercase mb-2">Step 1: Old Brain Signal (Neural)</label>

<button onclick="setSignal('UP')" id="btn-up" class="flex-1 py-2 rounded border border-slate-200 text-sm font-medium hover:bg-slate-50 transition active-signal ring-2 ring-emerald-500 bg-emerald-50 text-emerald-700">UP (Buy)</button>

<button onclick="setSignal('DOWN')" id="btn-down" class="flex-1 py-2 rounded border border-slate-200 text-sm font-medium hover:bg-slate-50 text-slate-600 transition">DOWN (Sell)</button>

<!-- Hurst Exponent -->

<label class="text-xs font-bold text-slate-500 uppercase">Hurst Exponent (Trend Power)</label>

0.65

```
                <input type="range" id="hurst-slider" min="0" max="1" step="0.05" value="0.65" class="w-full h-2 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-indigo-600">
                <p class="text-xs text-slate-400 mt-1"> > 0.5 = Trend, < 0.5 = Mean Reversion</p>
            </div>

            <!-- Entropy -->
            <div>
                <div class="flex justify-between mb-1">
                    <label class="text-xs font-bold text-slate-500 uppercase">Entropy (Chaos Level)</label>
                    <span id="entropy-val" class="text-xs font-mono font-bold text-rose-600">0.40</span>
                </div>
                <input type="range" id="entropy-slider" min="0" max="1" step="0.05" value="0.40" class="w-full h-2 bg-slate-200 rounded-lg appearance-none cursor-pointer accent-rose-600">
                <p class="text-xs text-slate-400 mt-1">Low (< 0.5) = Clean Market, High = Chaos</p>
            </div>

            <!-- Result Box -->
            <div id="logic-result" class="mt-6 p-4 rounded-lg bg-slate-900 text-center transition-all duration-300 transform scale-100">
                <p class="text-slate-400 text-xs uppercase tracking-widest mb-1">FINAL VERDICT</p>
                <h2 id="verdict-text" class="text-2xl font-black text-emerald-400">SNIPER BUY (CE)</h2>
                <p id="confidence-text" class="text-xs text-emerald-200 mt-1">Confidence: 95% (மரண மாஸ் என்ட்ரி)</p>
            </div>
```

```
            </div>
          </div>
        </div>

        <!-- Terminal Log -->
        <div class="bg-white rounded-xl border border-slate-200 shadow-sm overflow-hidden flex flex-col h-64">
          <div class="bg-slate-900 px-4 py-3 flex justify-between items-center">
            <h3 class="text-xs font-mono font-bold text-emerald-400 flex items-center gap-2">
              <span class="w-2 h-2 rounded-full bg-emerald-500 animate-pulse"></span> TERMINAL_LOG.exe
            </h3>
          </div>
          <div id="console-output" class="console-log flex-1 bg-slate-950 p-4 font-mono text-xs text-slate-300 overflow-y-auto space-y-2">
            <!-- Logs will appear here -->
          </div>
        </div>

      </div>

      <!-- CENTER COLUMN: Visualizations -->
      <div class="lg:col-span-8">

        <div class="bg-white rounded-xl border border-slate-200 shadow-sm">
          <!-- Tabs -->
          <div class="border-b border-slate-200">
```

```
<nav class="flex -mb-px" aria-label="Tabs">
  <button onclick="switchTab('kalman')" id="tab-kalman" class="w-1/3 py-4 px-1 text-center border-b-2 border-indigo-500 text-indigo-600 text-sm font-medium hover:text-indigo-800 transition">
    Rocket Kalman Filter
  </button>
  <button onclick="switchTab('indicators')" id="tab-indicators" class="w-1/3 py-4 px-1 text-center border-b-2 border-transparent text-slate-500 text-sm font-medium hover:text-slate-700 hover:border-slate-300 transition">
    Smart Indicators (CPR)
  </button>
  <button onclick="switchTab('fisher')" id="tab-fisher" class="w-1/3 py-4 px-1 text-center border-b-2 border-transparent text-slate-500 text-sm font-medium hover:text-slate-700 hover:border-slate-300 transition">
    Sniper Fisher Reversal
  </button>
</nav>
</div>

<div class="p-6">

<!-- VIEW 1: KALMAN FILTER -->
<div id="view-kalman" class="block">
  <div class="flex justify-between items-start mb-4">
    <div>
      <h4 class="font-bold text-lg text-slate-800">Kalman Trend Tracking</h4>
      <p class="text-sm text-slate-500">"Missile-Grade Trend Tracking Activated." விலை Kalman Line-க்கு மேல இருந்தா அப் ட்ரெண்ட்.</p>
    </div>
```

```
            <div class="flex items-center gap-2">
              <label class="text-xs text-slate-500">Gain (Sensitivity):</label>
              <select id="kalman-gain" class="text-xs border-slate-300 rounded
focus:ring-indigo-500">
                <option value="low">Low (Smooth)</option>
                <option value="med" selected>Medium</option>
                <option value="high">High (Reactive)</option>
              </select>
            </div>
          </div>
          <div class="chart-container">
            <canvas id="kalmanChart"></canvas>
          </div>
          <div class="mt-4 grid grid-cols-2 gap-4 text-xs">
            <div class="bg-indigo-50 p-3 rounded text-indigo-800">
              <strong>Logic:</strong> Filters out market noise. If Price > Kalman
Line = BUY.
            </div>
            <div class="bg-slate-50 p-3 rounded text-slate-600">
              <strong>Status:</strong> Tracking Velocity...
            </div>
          </div>
        </div>

        <!-- VIEW 2: SMART INDICATORS -->
        <div id="view-indicators" class="hidden">
          <div class="flex justify-between items-start mb-4">
            <div>
```

```
                <h4 class="font-bold text-lg text-slate-800">Precision Core (CPR & Supertrend)</h4>
                <p class="text-sm text-slate-500">"சாதாரணமா இல்லாம, இது Trap Zone-ஐயும் கணிக்கும்."</p>
              </div>
            </div>
            <div class="chart-container">
              <canvas id="indicatorsChart"></canvas>
            </div>
             <div class="mt-4 flex gap-4 text-xs">
              <div class="flex items-center gap-2">
                <span class="w-3 h-3 rounded-full bg-blue-400 opacity-50"></span> CPR Zone (Support)
              </div>
              <div class="flex items-center gap-2">
                <span class="w-3 h-1 bg-green-500"></span> Supertrend (Buy)
              </div>
              <div class="flex items-center gap-2">
                <span class="w-3 h-1 bg-red-500"></span> Supertrend (Sell)
              </div>
            </div>
          </div>

          <!-- VIEW 3: FISHER TRANSFORM -->
          <div id="view-fisher" class="hidden">
            <div class="flex justify-between items-start mb-4">
              <div>
                <h4 class="font-bold text-lg text-slate-800">Ehlers Fisher Transform</h4>
```

```
                <p class="text-sm text-slate-500">"Targeting Exact Reversal Points." (Reversal Sniper)</p>
              </div>
            </div>
            <div class="chart-container">
              <canvas id="fisherChart"></canvas>
            </div>
            <div class="mt-4 bg-yellow-50 p-3 rounded text-yellow-800 text-xs border border-yellow-200">
              <strong>Strategy:</strong> Extremes at +2 or -2 indicate sharp reversals. Crosses of the signal line confirm the entry.
            </div>
          </div>


        </div>
      </div>


      <!-- Technical Health Panel -->
      <div class="mt-6 grid grid-cols-1 md:grid-cols-2 gap-6">
        <div class="bg-white p-6 rounded-xl border border-slate-200 shadow-sm">
          <h4 class="font-bold text-slate-800 mb-4 flex items-center gap-2">
            <i class="ph ph-gauge text-slate-500"></i> Technical Sniper (Health)
          </h4>
          <div class="space-y-4">
            <div>
              <div class="flex justify-between text-xs mb-1">
                <span class="text-slate-600">ADX (Trend Strength)</span>
                <span class="font-bold text-emerald-600">32 (Strong)</span>
              </div>
```

ADX 25க்கு மேல இருந்தாதான் டிரெண்ட் ஸ்ட்ராங்.

Connors RSI (Sniper)

15 (Oversold)

இது சாதாரண RSI விட பாஸ்டாரியாக்ட் பண்ணும்.

<h4 class="font-bold text-slate-800 text-sm">Security Lock Activated</h4>

பாஸ்வேர்ட் பாதுகாப்பு (Active)

```
                <button class="px-4 py-2 bg-slate-900 text-white text-xs font-bold rounded hover:bg-slate-700 transition">
                  ACCESS ADMIN PANEL
                </button>
              </div>
            </div>

        </div>
      </div>
    </div>

    <!-- JavaScript Application Logic -->
    <script>
      // --- DATA & STATE MANAGEMENT ---
      const state = {
        currentSignal: 'UP',
        hurst: 0.65,
        entropy: 0.40,
        kalmanGain: 'med',
        activeTab: 'kalman'
      };

      // --- CHART INSTANCES ---
      let kalmanChart, indicatorsChart, fisherChart;

      // --- HELPER: GENERATE MOCK DATA ---
      function generateData(points, type = 'trend') {
        let data = [];
```

```
  let price = 100;
  for (let i = 0; i < points; i++) {
    let noise = (Math.random() - 0.5) * 2; // -1 to 1


    if (type === 'trend') {
      // Trend with noise
      price += Math.sin(i / 10) * 0.5 + noise + 0.2;
    } else if (type === 'oscillator') {
      // Sine wave for Fisher
      price = Math.sin(i / 5) * 1.5 + (Math.random() * 0.5);
    } else if (type === 'candles') {
      // OHLC mock
      let change = (Math.random() - 0.45) * 2;
      let close = price + change;
      let high = Math.max(price, close) + Math.random();
      let low = Math.min(price, close) - Math.random();
      data.push({ x: i, o: price, h: high, l: low, c: close });
      price = close;
      continue;
    }
    data.push(price);
  }
  return data;
}


// --- INITIALIZATION ---
document.addEventListener('DOMContentLoaded', () => {
  initCharts();
```

```
  updateLogicEngine();
  startLogStream();
  setupEventListeners();
});

function setupEventListeners() {
  document.getElementById('hurst-slider').addEventListener('input', (e) => {
    state.hurst = parseFloat(e.target.value);
    document.getElementById('hurst-val').innerText = state.hurst.toFixed(2);
    updateLogicEngine();
  });

  document.getElementById('entropy-slider').addEventListener('input', (e) => {
    state.entropy = parseFloat(e.target.value);
    document.getElementById('entropy-val').innerText = state.entropy.toFixed(2);
    updateLogicEngine();
  });

  document.getElementById('kalman-gain').addEventListener('change', (e) => {
    state.kalmanGain = e.target.value;
    updateKalmanChart(); // Re-render logic
  });
}

// --- 1. THE LOGIC ENGINE (NEURO-QUANTUM CORE) ---
function setSignal(signal) {
  state.currentSignal = signal;
```

```
    // Update UI Buttons
    const btnUp = document.getElementById('btn-up');
    const btnDown = document.getElementById('btn-down');

    if(signal === 'UP') {
      btnUp.className = "flex-1 py-2 rounded border border-slate-200 text-sm font-medium hover:bg-slate-50 transition active-signal ring-2 ring-emerald-500 bg-emerald-50 text-emerald-700";
      btnDown.className = "flex-1 py-2 rounded border border-slate-200 text-sm font-medium hover:bg-slate-50 text-slate-600 transition";
    } else {
      btnDown.class Name = "flex-1 py-2 rounded border border-slate-200 text-sm font-medium hover:bg-slate-50 transition active-signal ring-2 ring-rose-500 bg-rose-50 text-rose-700";
      btnUp.class Name = "flex-1 py-2 rounded border border-slate-200 text-sm font-medium hover:bg-slate-50 text-slate-600 transition";
    }
    updateLogicEngine();
  }

  function updateLogicEngine() {
    // Implementation of the "SECRET FORMULA" from the PDF
    const { currentSignal, hurst, entropy } = state;
    const resultBox = document.getElementById('logic-result');
    const verdictText = document.getElementById('verdict-text');
    const confText = document.getElementById('confidence-text');

    let verdict = "WAIT";
    let colorClass = "text-slate-400";
    let bgClass = "bg-slate-900";
```

```
let confidence = 0;
let tamilMsg = "";

// TRAP CHECK FIRST
if (entropy > 0.8) {
  verdict = "TRAP DETECTED";
  colorClass = "text-slate-300";
  bgClass = "bg-slate-700";
  tamilMsg = "Warning: Chaos High! (ஆபத்து)";
  confidence = 0;
} else {
  if (currentSignal === "UP") {
    if (hurst > 0.6 && entropy < 0.5) {
      verdict = "SNIPER BUY (CE)";
      colorClass = "text-emerald-400";
      bgClass = "bg-emerald-950";
      confidence = 95;
      tamilMsg = "மரண மாஸ் என்ட்ரி (Confirmed)";
    } else if (hurst > 0.5) {
      verdict = "BUY (CE)";
      colorClass = "text-emerald-400";
      bgClass = "bg-emerald-900";
      confidence = 70;
      tamilMsg = "Normal Buy Setup";
    }
  } else {
    if (hurst > 0.6 && entropy < 0.5) {
      verdict = "SNIPER SELL (PE)";
```

```
            colorClass = "text-rose-400";
            bgClass = "bg-rose-950";
            confidence = 95;
            tamilMsg = "மரண மாஸ் என்ட்ரி (Confirmed)";
          } else if (hurst > 0.5) {
            verdict = "SELL (PE)";
            colorClass = "text-rose-400";
            bgClass = "bg-rose-900";
            confidence = 70;
            tamilMsg = "Normal Sell Setup";
          }
        }
      }

      // UI Update
      verdictText.innerText = verdict;
      verdictText.className = `text-2xl font-black ${colorClass}`;
      resultBox.className = `mt-6 p-4 rounded-lg text-center transition-all duration-
300 transform scale-100 ${bgClass}`;
      confText.innerText = `Confidence: ${confidence}% (${tamilMsg})`;
    }

    // --- 2. TERMINAL LOG ---
    function startLogStream() {
      const logs = [
        "[AI] Analyzing Phase Shift in 5m timeframe...",
        "[DATA] Put Call Ratio: 1.2 (Bullish)",
        "[ALERT] Volatility Spike detected in Bank Nifty",
```

```
    "[SYSTEM] Adjusting Trailing Stoploss +5 pts",
    "[CHECK] Order flow from FII/DII is stable",
    "[SCAN] No Trap detected in current zone",
    "[NEURAL] LSTM Layer 2 Activation: 0.89",
    "[FISHER] Reversal imminent on 15m chart"
  ];

  const container = document.getElementById('console-output');

  setInterval(() => {
    const msg = logs[Math.floor(Math.random() * logs.length)];
    const time = new Date().toLocaleTimeString('en-US', { hour12: false });

    const line = document.createElement('div');
    line.innerHTML = `<span class="text-slate-500">[${time}]</span> ${msg}`;
    container.appendChild(line);
    container.scrollTop = container.scrollHeight;

    if(container.children.length > 20) container.removeChild(container.firstChild);
  }, 2500);
}

// --- 3. CHART VISUALIZATIONS ---

// Tab Switcher
window.switchTab = function(tabName) {
  // Hide all
  document.getElementById('view-kalman').classList.add('hidden');
```

```
    document.getElementById('view-indicators').classList.add('hidden');
    document.getElementById('view-fisher').classList.add('hidden');

    // Reset styles
    ['kalman', 'indicators', 'fisher'].forEach(t => {
      const btn = document.getElementById(`tab-${t}`);
      btn.className = "w-1/3 py-4 px-1 text-center border-b-2 border-transparent text-slate-500 text-sm font-medium hover:text-slate-700 hover:border-slate-300 transition";
    });

    // Show active
    document.getElementById(`view-${tabName}`).classList.remove('hidden');
    document.getElementById(`tab-${tabName}`).className = "w-1/3 py-4 px-1 text-center border-b-2 border-indigo-500 text-indigo-600 text-sm font-medium hover:text-indigo-800 transition";
  };

  function initCharts() {
    // -- KALMAN CHART --
    const kalmanCtx = document.getElementById('kalmanChart').getContext('2d');
    const kDataRaw = generateData(60, 'trend');
    // Simple Moving Average as mock Kalman for visual
    const kLine = kDataRaw.map((v, i) => i < 5 ? v : (v + kDataRaw[i-1] + kDataRaw[i-2])/3);

    kalmanChart = new Chart(kalmanCtx, {
      type: 'line',
      data: {
```

```
    labels: Array.from({length: 60}, (_, i) => i),
    datasets: [
      {
        label: 'Price',
        data: kDataRaw,
        borderColor: '#cbd5e1',
        borderWidth: 1,
        pointRadius: 0,
        tension: 0.1
      },
      {
        label: 'Kalman Trend',
        data: kLine,
        borderColor: '#4f46e5',
        borderWidth: 3,
        pointRadius: 0,
        tension: 0.4
      }
    ]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    plugins: { legend: { display: true } },
    scales: { x: { display: false }, y: { display: true } }
  }
});
```

```
// -- INDICATORS CHART --
const indCtx = document.getElementById('indicatorsChart').getContext('2d');
// Mock OHLC for visual representation in line form for simplicity in this demo,
// but representing the "Bands"
const pData = generateData(60, 'trend');
const upperBand = pData.map(v => v + 2);
const lowerBand = pData.map(v => v - 2);

indicatorsChart = new Chart(indCtx, {
  type: 'line',
  data: {
    labels: Array.from({length: 60}, (_, i) => i),
    datasets: [
      {
        label: 'Supertrend Upper',
        data: upperBand,
        borderColor: 'rgba(16, 185, 129, 0.3)', // Green transparent
        backgroundColor: 'rgba(16, 185, 129, 0.1)',
        fill: '+1',
        pointRadius: 0,
        borderWidth: 1
      },
      {
        label: 'Supertrend Lower',
        data: lowerBand,
        borderColor: 'rgba(244, 63, 94, 0.3)', // Red transparent
        pointRadius: 0,
        borderWidth: 1
```

```
        },
        {
          label: 'Close Price',
          data: pData,
          borderColor: '#334155',
          borderWidth: 2,
          pointRadius: 1
        }
      ]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      plugins: { legend: { display: false } },
      scales: { x: { display: false } }
    }
});

// -- FISHER CHART --
const fisherCtx = document.getElementById('fisherChart').getContext('2d');
const fData = generateData(60, 'oscillator');
const trigger = fData.map((v, i) => i > 0 ? fData[i-1] : 0);

fisherChart = new Chart(fisherCtx, {
    type: 'line',
    data: {
      labels: Array.from({length: 60}, (_, i) => i),
      datasets: [
```

```
      {
        label: 'Fisher',
        data: fData,
        borderColor: '#0ea5e9',
        borderWidth: 2,
        tension: 0.4,
        pointRadius: 0
      },
      {
        label: 'Trigger',
        data: trigger,
        borderColor: '#94a3b8',
        borderWidth: 1,
        borderDash: [5, 5],
        pointRadius: 0
      }
    ]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    scales: {
      x: { display: false },
      y: {
        min: -3, max: 3,
        grid: { color: (ctx) => (ctx.tick.value === 0 ? '#64748b' : '#e2e8f0') }
      }
    }
```

```
        }
      });
    }

    function updateKalmanChart() {
      // Simulate changing gain by smoothing the line more or less
      const kDataRaw = kalmanChart.data.datasets[0].data;
      let smoothFactor = 3;
      if(state.kalmanGain === 'low') smoothFactor = 8;
      if(state.kalmanGain === 'high') smoothFactor = 1;

      const newKLine = kDataRaw.map((v, i) => {
        if(i < smoothFactor) return v;
        let sum = 0;
        for(let j=0; j<smoothFactor; j++) sum += kDataRaw[i-j];
        return sum/smoothFactor;
      });

      kalmanChart.data.datasets[1].data = newKLine;
      kalmanChart.update();
    }

  </script>
</body>
</html>
```